

Luca Drole

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Education

- Sept 2023 **M.Sc. Biomedical Engineering - Biomedical Imaging Major, ETH Zürich**
○ Excellence Scholar 2023 (ESOP)
- Sept 2020 – **B.Sc. Biomedical Engineering, Polytechnic University of Milan, 110 cum Laude**
Jul 2023 ○ Course Representative, Student Research at NECSTLab.
- 2015–2020 **Scientific Lyceum, L.S. Oberdan, Trieste, 100 cum Laude**

Research Experience

- Oct 2025 **Master's Thesis Student, BMIC Group, ETH Zürich**
Data-efficient architectures for Vision Transformers in Medical Imaging. Supervised by Dr. Ertunc Erdil.
- Mar 2025 – **Visiting Student, A. Martinos Center for Biomedical Imaging, Massachusetts**
Sept 2025 **General Hospital & Harvard Medical School**
Working on contrast-agnostic MRI feature extraction at the LEMOn group under the supervision of Dr. Peirong Liu and Dr. Xiaoling Hu.
- Feb 2024 – **Semester Project, CMR Group, ETH Zürich**
Jun 2024 Title: "*Unsupervised Image to Scar Segmentation Transformation using a CycleGAN*"
Worked with a CycleGAN paradigm to automate cardiac scar segmentation, under the supervision of I. Margolis and Dr. L. Dal Toso.
- Feb 2023 – **Bachelor's Thesis Student, NECSTLab, PoliMi**
Jul 2023 Title: "*Segmentation of cell nuclei from histopathological images*". Developed a novel loss function for the Hover-Net architecture to compensate for class imbalance. Worked with PyTorch and other popular libraries. Supervised by I. Poles and Dr. E. D'Arnese. [github](#)
- Nov 2022 – **Research Intern, Polyclinic Hospital Milan - Polytechnic University of Milan**
Jun 2023 Multimodal analysis of EEG-fMRI data from patients with Major Depressive Disorder. Supervised by Professor E. Maggioni.
- Jul 2022– **Amgen Scholar - Research Intern, ETH Zürich**
Sept 2022 Fully funded research internship at the Laboratory for Movement Biomechanics. Automated pose estimation of joint implants from moving fluoroscopy videos, employing image processing techniques and deep learning (PyTorch). Supervised by Dr. Florian Vogl. Poster presented at the *Amgen Scholars European Symposium* (University of Cambridge, 2022).

Conference Papers

- [1] Luca Drole, Isabella Poles, Eleonora D'Arnese, and Marco D Santambrogio. Towards a lightweight 2D U-Net for accurate semantic segmentation of kidney tumors in abdominal CT images. In *IEEE EUROCON 2023 International Conference on Smart Technologies*, 2023.

Teaching & Leadership Experience

- Sept 2024 **Teaching Assistant, ETH Zürich**
TA for the "Programming for Engineers" and "Biomedical Imaging" courses (Autumn Semester 2024 and 2025). Responsible for grading, presentations, student support, and updating the lecture material.
- Oct 2020 – **Junior Board Member for Fundraising, European Youth Parliament Italy**
Nov 2022 EYP Italy is a youth-led organisation that involves more than 800 students yearly in educational events.
○ Involved in the association's governance, managed relations with partners and patrons, grants applications, supported event planning and on-site management.
- Jan 2021– **Project Manager, European Youth Parliament Italy, Trieste**
Nov 2021 Organized a four-day simulation of the European Parliament, with the theme: "Rethinking the future of Europe: Unity for a More Integrated Society"

Awards

- March 2025 **Nova Italy 111 Students List**
Ranked among the top 10 Italian students for the "Other Engineering & Technology" category.
- Sept 2023 **Excellence Scholarship, ETH Zürich**
Awarded a CHF 48 000 scholarship to pursue Master's studies.
- Sept 2022 **Runner-up Best Poster Prize, Amgen Scholars European Symposium, Cambridge University**
- Jan 2022 **Best Freshmen Award, Polytechnic University of Milan**
- Aug 2020 **National Excellence Honours Roll, Italian Ministry of Education**

Volunteering and Organisations

- Apr 2018 – **European Youth Parliament**
Present Experienced team leader, academic facilitator, and facilitator trainer. I directed several policy discussion groups on various topics related to EU legislation. Relevant focus on EU Health, Tech, and Energy policy. Volunteered at around 30 educational events across Europe.

Certifications

IELTS Academic 8.5

Computer Skills

- Proficient Python, PyTorch
Intermediate Tensorflow, Matlab, Git, L^AT_EX

Languages

- Native Italian
C2 English
A2 German